

**School of Computer Applications and Technology**

Mid Term Examination - Apr 2026

**M.Sc. CS II Sem / MCA (All) II Sem / Int. BCA+MCA VIII Sem**Duration : 90 Minutes  
Max Marks : 50**Sem II - E1PY206T - Data Communication and Networking**General Instructions*Answer to the specific question asked**Draw neat, labelled diagrams wherever necessary**Approved data hand books are allowed subject to verification by the Invigilator*

- |    |                                                                                                                           |         |
|----|---------------------------------------------------------------------------------------------------------------------------|---------|
| 1) | Define the working of Gateway in a Computer Network.                                                                      | CO1(3)  |
| 2) | Explain the advantages of star topology.                                                                                  | CO1(5)  |
| 3) | Differentiate between simplex, half duplex and full duplex with suitable examples.                                        | CO2(5)  |
| 4) | Explain any two types of Unguided Media.                                                                                  | CO1(8)  |
| 5) | Apply any two encoding techniques to represent given data signal: 11010.                                                  | CO2(8)  |
| 6) | Analyze how the criteria of reliability, performance, and security influence the design of a computer network.            | CO1(9)  |
| 7) | Evaluate the different types of computer networks (LAN, MAN, WAN) and justify their suitability for various applications. | CO1(12) |

**OR**

- |                                                                                                                                                                      |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| A communication system uses Pulse Code Modulation to transmit analog signals. Evaluate the complete PCM process (sampling, quantization, and encoding) with example. | CO2(12) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|